



Analyzing Sentiments of Fine Food Reviews on Amazon

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COMPSCI 715 Programming for ML

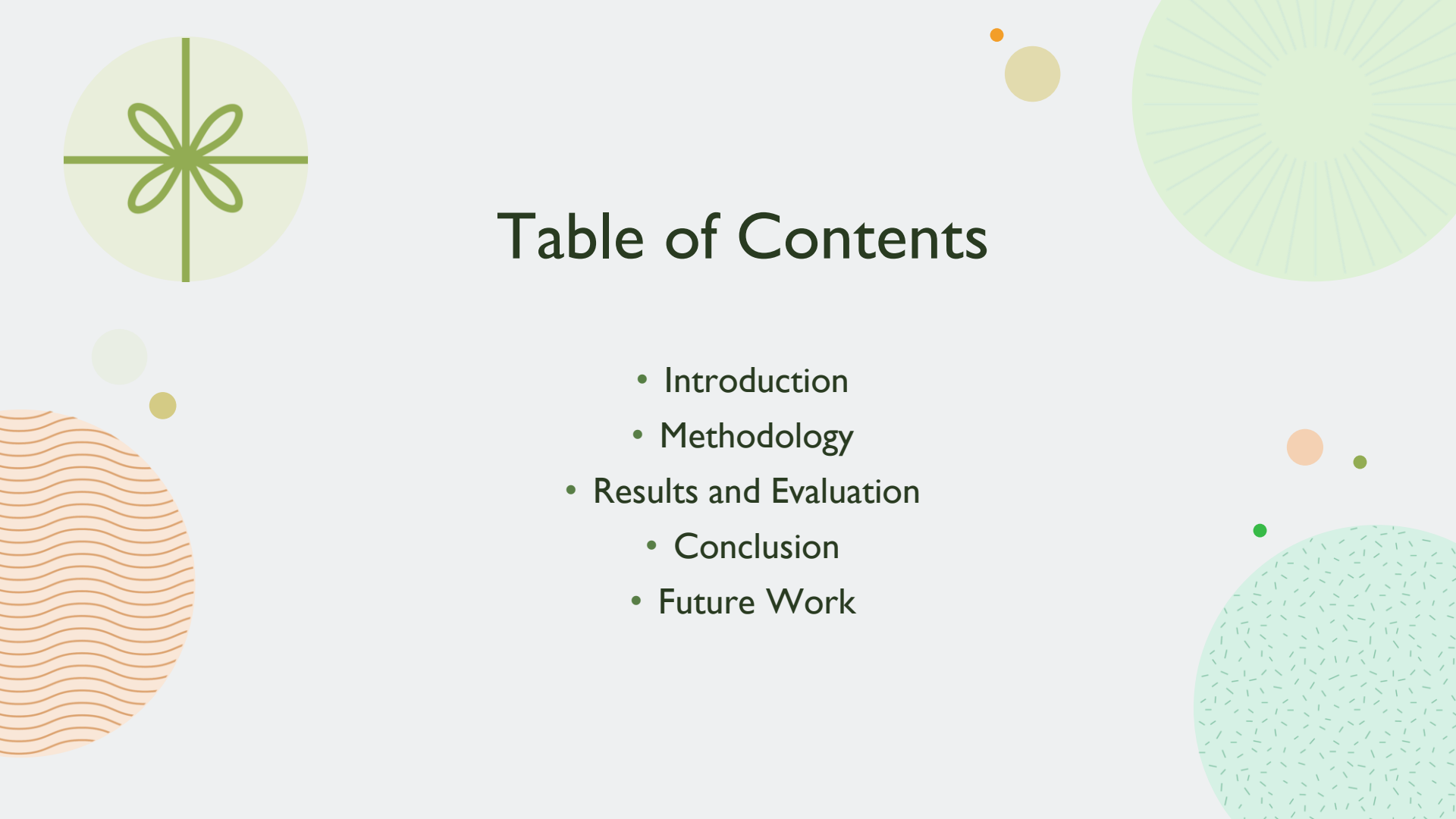


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Introduction

What?

- Analyze customer sentiments on Amazon fine food reviews using VADER, Logistic Regression and DistilBERT Sentiment Model.

Why?

- To understand customer feedback for product improvements
- Performing sentiment analysis will help product owners understand customer satisfaction and dissatisfaction

How?

- Use **VADER** Model.
- Use **Logistic Regression** Model.
- Use **DistilBERT** Sentiment Model.
- Evaluate and compare the results by all three models.

Methodology



Step 1: Data
Collection



Step 2: Data pre-
processing



Step 3: Basic
Exploratory Data
Analysis



Step 4: Applying
VADER Model



Step 5: Applying
Logistic Regression
Model



Step 6: Applying
DistilBERT Sentiment
Model



Step 7: Evaluating
Results

VADER Model

- VADER (Valence Aware Dictionary and sEntiment Reasoner) is a rule-based sentiment analysis tool specifically designed to analyze social media texts.
- It is a pre-trained sentiment analysis model that provides sentiment scores for given text
- VADER has 4 categories positive, negative, neutral and compound
- Compound is a calculation of positive, negative and neutral score ranging between -1 to 1
- VADER is easily available as a part of the NLTK (Natural Language Toolkit) library in python.

Results by VADER Sentiment Analysis

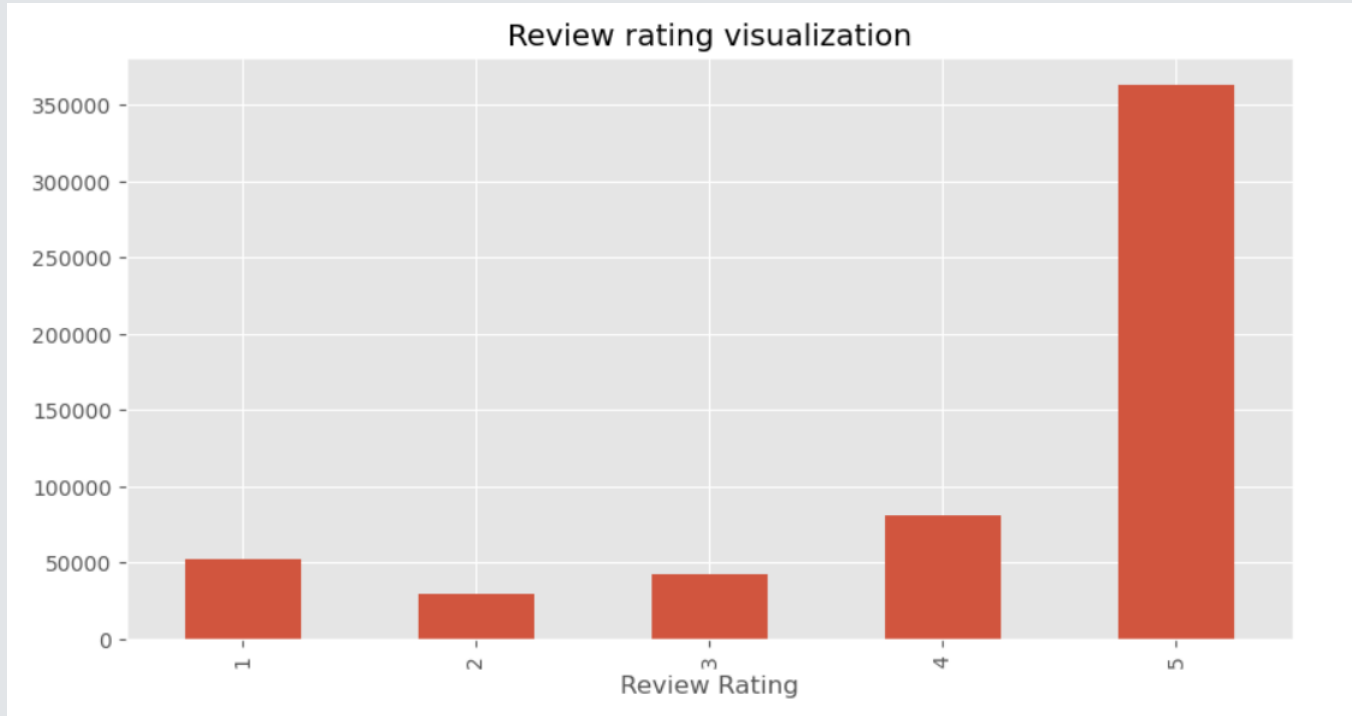


Figure 1: Exploratory Data Analysis

Results by VADER Sentiment Analysis

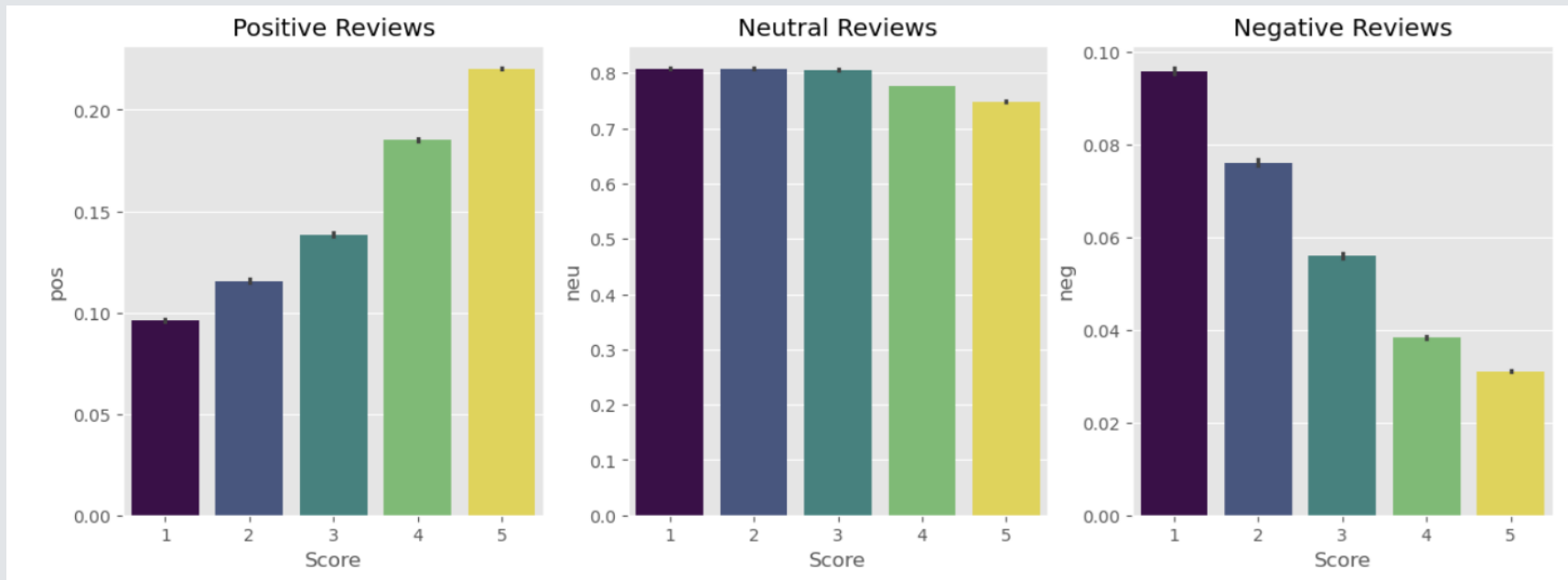


Figure 2: Customer Rating vs VADER Sentiment Score

Results by VADER Sentiment Analysis

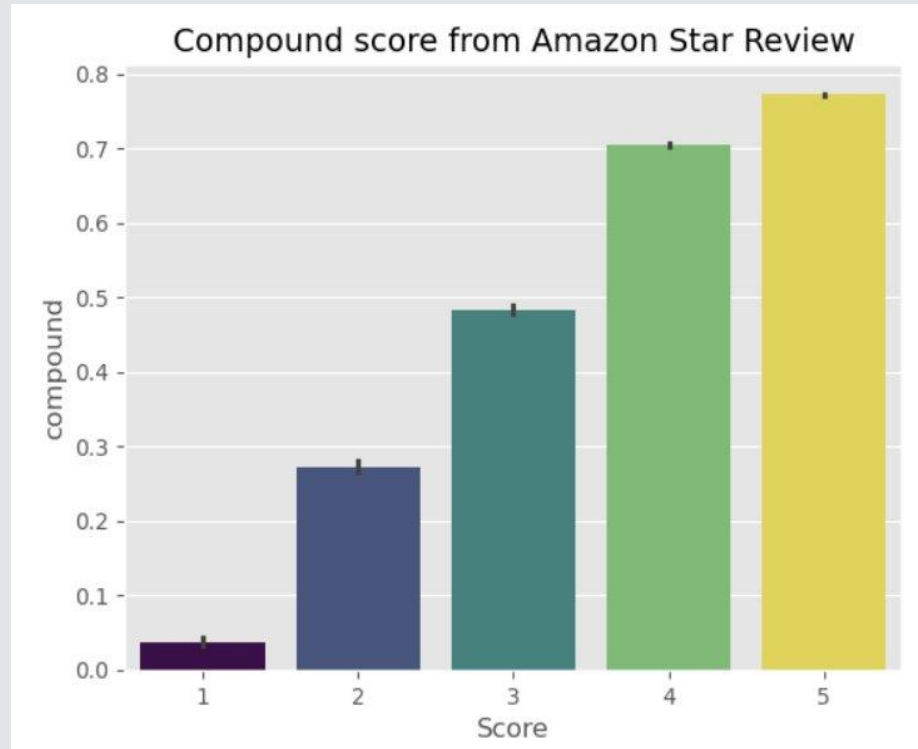


Figure 3: Customer Rating vs VADER Compound score

Logistic Regression Model for Sentiment Analysis

- Logistic Regression is a supervised machine learning model used for classification.
- Review text is first converted into numerical features using TF-IDF vectorization.
- TF-IDF gives higher importance to words that are meaningful in a review but not too common across all reviews.
- The model learns patterns between review text and sentiment labels.
- In this project, ratings were converted into sentiment classes, and Logistic Regression was trained to predict whether a review is positive or negative.
- It works as a strong baseline model because it is fast, interpretable, and performs well on text classification tasks.

Results by Logistic Regression Model

- Logistic Regression achieved approximately 93% accuracy on the test data.
- The model performed very well on positive reviews, with high precision and recall.
- However, negative reviews were harder to classify compared to positive reviews.
- This may be due to the dataset containing higher positive reviews when compared to negative reviews.
- Logistic Regression served as a strong baseline model before comparing with transformer-based BERT analysis.

```
Classification Report:
              precision    recall  f1-score   support

     0.0         0.84      0.68      0.75     16407
     1.0         0.94      0.98      0.96     88756

 accuracy          0.93     105163
 macro avg         0.89      0.83      0.86     105163
 weighted avg      0.93      0.93      0.93     105163
```

```
Confusion Matrix:
[[11232  5175]
 [ 2209 86547]]
Accuracy Score: 0.9297851906088643
```

Top 10 Highest and Lowest Sentiment Products using Logistic Regression Model

Top 10 Performing Products (Based on Sentiment):

	ProductId	sentiment \
156	B0036VM05I	1.0
35633	B0036UQTQA	1.0
36332	B0036UWPJ0	1.0
105943	B0036VE3V2	1.0
228469	B0036V7K4E	1.0
229039	B0036VFX5M	1.0
309833	B0036UV8RU	1.0
487525	B001XWR35Y	1.0
539149	B0036VFTMO	1.0
544019	B0036UUY9I	1.0

Summary

156	Great Deal
35633	New label, same great stuff
36332	Works really well
105943	My Dog Thinks They Are The Best!
228469	You can't buy cheddar like this at the grocery...
229039	Not for everyone but I LOVE it!
309833	I love the coffee, but:
487525	Best Risotto in the World
539149	Dog + cats LOVE it!
544019	YUMMMMMY

Worst 10 Performing Products (Based on Sentiment):

	ProductId	sentiment \
93743	B00514BIQK	0.0
173990	B005116LQU	0.0
210927	B0025VF8TK	0.0
226725	B0025UOODC	0.0
312697	B004TMAAM8	0.0
369755	B0015DEH8C	0.0
369936	B0015DEH8W	0.0
396416	B0025VP9I0	0.0
456925	B005116KK2	0.0
525741	B005116LMO	0.0

Summary

93743	WARNING to SENSITIVE Celiacs!!
173990	Lowell Foods Dew Sugar
210927	Doesn't look like a salmon, doesn't taste like...
226725	Terrible After Taste
312697	Harm no Charm
369755	Do not order... stale candy!
369936	Sent an open package containing only 28 pieces
396416	item not as shown or name ordered
456925	Worst pork pate ever
525741	Disappointed with quality

BERT Sentiment Analyzer (DistilBERT sentiment model)

- BERT is a transformer-based language model that understands the meaning of words based on context.
- Unlike TF-IDF, BERT does not treat words as isolated tokens; it considers the full sentence structure.
- A pretrained DistilBERT sentiment model was used to classify review text as positive or negative.
- Since BERT was already trained on large text datasets, it was applied without fine-tuning due to CPU runtime limitations.
- BERT provides both a sentiment label and a confidence score for each review.
- This helps compare traditional sentiment methods with a modern transformer-based approach.

Note: Due to computational limitations DistilBERT was only applied on a sample dataset of 5000 rows.

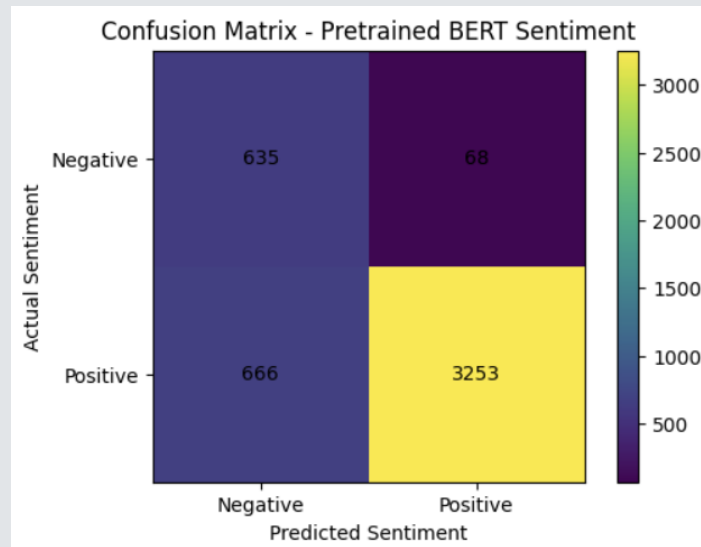
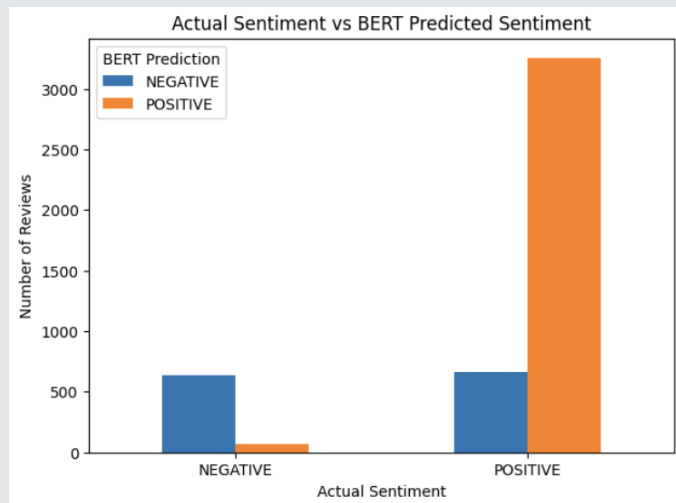
Results by DistilBERT sentiment model

Accuracy of DistilBERT Model: 0.8412

Evaluated rows: 4622

Positive F1-score: 0.90

Negative F1-score: 0.63



- Correctly identified most negative reviews (635 of 703)
- Confidence scores were highly concentrated near 1.0

Figure 4: Confusion Matrix, Actual vs Predicted sentiment scores

Results:

VADER Sentiment Analysis Accuracy:

```
Accuracy for VADER model: 0.7970
```

Logistic Regression Analysis Accuracy:

Classification Report:

	precision	recall	f1-score	support
0.0	0.84	0.68	0.75	16407
1.0	0.94	0.98	0.96	88756
accuracy			0.93	105163
macro avg	0.89	0.83	0.86	105163
weighted avg	0.93	0.93	0.93	105163

Confusion Matrix:

```
[[11232 5175]
```

```
[ 2209 86547]]
```

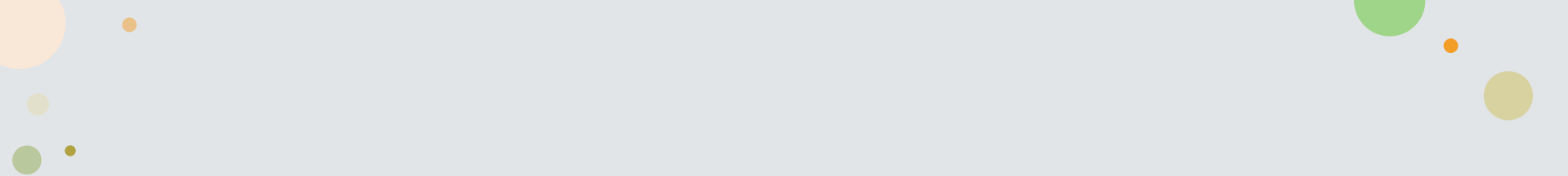
Accuracy Score: 0.9297851906088643

DistilBERT Model Analysis Accuracy:

BERT Accuracy: 0.8412

	precision	recall	f1-score	support
NEGATIVE	0.49	0.90	0.63	703
POSITIVE	0.98	0.83	0.90	3919
accuracy			0.84	4622
macro avg	0.73	0.87	0.77	4622
weighted avg	0.90	0.84	0.86	4622

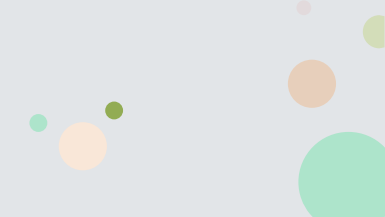
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array([[ 635,  68],  
       [ 666, 3253]])
```

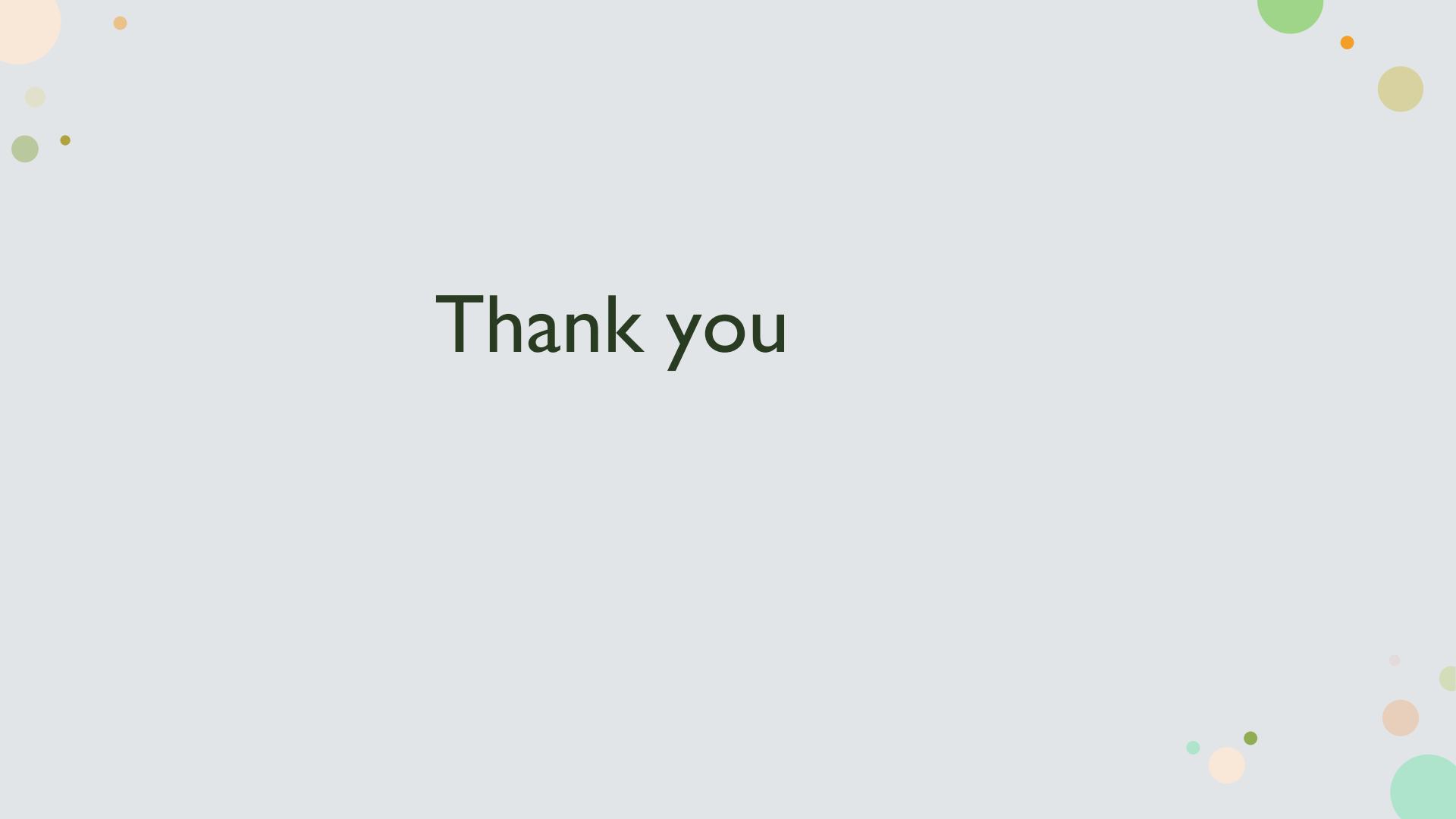


Conclusion and Future work

Which model performed better and what could be the reason?

How can this work be used in future for better sentiment analysis?



The slide features a light gray background with decorative elements in the corners. The top-left corner contains several small circles in orange, yellow, and green. The top-right corner features larger green and yellow circles, along with a small orange dot. The bottom-right corner is decorated with a cluster of circles in teal, orange, and light green. The text "Thank you" is centered in a dark green, sans-serif font.

Thank you